

Histology

1- Which cell has no nucleus?

- Normoblast
- Erythrocyte
- Basophil
- Neutrophil

2- What is the name of macrophage cells in liver?

- Osteoclast
- Langerhans
- Microglial
- Kupffer

3- Retrograde transportation in the axon is occurred by:

- Kinesin
- Actin
- Dynein
- Talin

4- An exocrine gland that loses cell bodies during secretion, as is the case for the sebaceous gland, is further classified as a.....gland.

- Apocrine
- Holocrine
- Merocrine
- Endocrine

5- In which type of granules in platelets we can find lysosomic enzyme?

- Alpha
- Delta
- Landa
- Beta

6- Which of the following histological structures has Chondroblast?

- Wolkman Canal
- Chondrium Matrix
- Endosteum
- Peroosteom

7- Structure of framework of spleen and lymphnodes are composed of which fibers?

- elastic
- elaunin
- collagen bundles
- reticular

8- What is the following is INCORRECT?

- cardiac muscle and skeletal muscle have sarcomere
- Striation was seen in transverse section of cardiac muscle and skeletal muscle
- Striation was seen in longitudinal section of cardiac muscle and skeletal muscle
- cardiac muscle and smooth muscle nucleus located in center part

9- About Unmyelinated PNS axons, which choice is correct?

- Oligodendrocyte cell starts to envelop multiple axons
- Schwann cell starts to envelop multiple axons
- Schwann cell starts to envelop one axon
- Oligodendrocyte cell starts to envelop one axon

10- Which layer of the epidermis has cells which have keratohyaline granules?

- Stratum basale
- Stratum spinosum
- Stratum corneum
- Stratum granulosum

11- Which of the following is a supporting cell found in the ganglia?

- Satellite cells
- Schwann cells
- Basket cells
- Neuroglia

12- All of these are located in the chemical synapses, except:

- Postsynaptic receptors
- Gap junctions
- Synaptic vesicles
- Mitochondria

13- Which cell has a part in formation of blood brain barrier?

- Astrocyte
- Ependymal cell
- Oligodendrocyte
- Schwann cell

14- Which of the following Is Observed in the histological structure Metaphysis Plate

- Conderoclast
- Chonderocyte
- Osteoclast
- Osteocyte

15- What is line that bisects the dark band in muscle?

- M line
- I band
- A band
- H band

16- What type of muscles have branching or spindle shape cells?

- Smooth- Striated
- Cardiac- Smooth
- Striated - Smooth
- Skeletal- Cardiac

17- A nerve is formed through a collection of

- Glial cells
- Myelin sheet
- Neurons
- Axones

18- Which layer of the epidermis contains cells with no nuclei or organelles?

- spinosum
- corneum
- granulosum
- basale

19- In which cell nucleus has cartwheel or clock face arrangement?

- Plasmacell
- Mastcell
- Fibrocyte
- Fibroblast

20- In which cell granules composed of major basic proteins?

- Basophil
- Lymphocyte
- Neutrophil
- Eosinophil

21- Which cell is the largest white blood cell?

- Neutrophil
- Monocyte
- Basophil
- Lymphocyte

22- Spinal central canal covered by:

- Oligodendrocyte
- Schwann cell
- Microglia
- Ependymal cell

23- Identify the INCORRECT statement

- In pseudostratified epithelia all cells are in contact with the basement membrane.
- The basement membrane consists of two sub layers: the reticular layer and the basal lamina.
- Epithelia are classified by the shape of cell layers
- Microvilli increase the surface area of the epithelia cells

24- In which of the following is Observed Osteogenic cell & Osteoclast ?

- Nothing
- Endosteum
- Periosteum
- Pericardium

25- Which of muscle protein has three subunit(TnT,TnC,TnI)?

- Myosin
- Troponin
- Actin
- Tropomyosin

26- Nissl bodies in the perikaryon of neurons are composed of:

SER and Lysosomes

Golgi and RER

Ribosomes and RER

Ribosomes and Lysosomes

27- In which of the following is Observed only type I collagen ?

Perichondrium

Hyalin Cartilage

Matrix

Elastic Cartilage

28- What do you call the simple squamous epithelium that lines the blood vessels?

Pseudostratified

Epithelioid tissue

Mesothelium

Endothelium

29- What is the covering of an individual muscle fiber?

Sarcolemma

Epimysium

Endomysium

Perimysium

30- In which tissue the nucleus are at the rim of the cells?

Fibroblast

Fibrocyte

Macrophage

Adipocyte

31- Which cell is sensory reception between epidermal cells?

- Merkel
- Lymphocyte
- Melanocyte
- Keratinocyte

32- Which cell secrete antibody in connective tissue?

- Fibrocyte
- Mastcell
- Plasmacell
- Adibocyte

33- What type of glands are the ceruminous glands in ear?

- Apocrine gland
- Eccrine gland
- Sebaceous glands
- Endocrine gland

34- Which types of neurons are located in dorsal root ganglia?

- Pseudounipolar
- Anaxonic
- Bipolar
- Multipolar

35- Which type of junctions is an effective barrier to the diffusion of substances across an epithelium?

- Gap junctions
- Tight junctions
- Hemidesmosomes
- Desmosomes

36- This type of neuron consist of many dendrites and one true axon , and are more:

Multipolar

Unipolar or pseudo unipolar

Bipolar

Apolar

Embryology

37- Implantation of the blastocyst

cannot occur without prior

differentiation of syncytiotrophoblast

.hatching from the zona pellucida

.fragmentation of the primary yolk sac

development of amnioblast

Embryology

38- Major events beginning in the second week of development include:

notochord differentiation

somite formation

angiogenesis

yolk sac development

39- Somites

are the source of cells in the nucleus pulposus

give rise to cardiac muscle

give rise to the cartilages of the branchial arches

are derived from the paraxial mesoderm

40- The outer layer of the umbilical cord is cont

inuuous with, and similar in structure to the

chorionic membrane

amniotic membrane

syncytiotrophoblast

cytotrophoblast

41- Implantation is preceded by an increase in ovarian hCG secretion

usually occurs near the end of the first week following fertilization

requires contact of the zona pellucida with uterine epithelium

cannot occur outside the uterus and uterine tubes

leads to the development of the inner cell mass

42- The primitive streak

is derived from epiblast

first appears in the roof of the amniotic cavity

gives rise to ectomesenchyme of the 1st branchial arch

gives rise to the allantois

43- Which layer of the bilaminar embryonic disc forms part of the lining of the amniotic cavity?

epiblast

hypoblast

blastocyst

trophoblast

44- Pregnancy tests generally are based upon the detection in maternal urine of:

- Estrogen
- Progesterone
- human Chorionic Somatomotropin
- human Chorionic Gonadotrophin

45- The intraembryonic coelom

- it is lined by endoderm
- normally contains only fluid
- becomes the pleural and peritonea
- , but not the pericardial, cavities
- all of the above

46- 21 Mesoderm lateral to the notochord:

- is derived from hypoblast
- differentiates into somite
- migrates cranial to prochordal plate
- induces endoderm differentiation

47- As the follicle grows, the oogonium becomes located in a mound of follicle cells called the:

- vesicular antrum
- oocyte
- cumulus oophorus
- Mural granulosa cells

48- Which of the following is associated with oligohydramnios

- tracheo-esophageal fistula
- . renal agenesis
- myelomeningocele
- anencephal

49- Extraembryonic somatic mesoderm is in close association with:

- syncytiotrophoblast
- cytotrophoblast & amnion
- yolk sac
- syncytio- & cytotrophoblast

50- A 3 year-old male patient presents with epicanthal folds, flat nasal bridge, single palmar crease. The patient is 47 XY on karyotype. What could be the cause of this karyotype?

- Chromosomal mutation
- . Chromosomal inversion
- Chromosomal nondisjunction
- Chromosomal deletion

51- Neural crest

- gives rise to most of the cardiogenic mesoderm
- is derived from the epiblast
- .becomes smooth muscle cells within the wall of the bowel
- .induces development of syncytiotrophoblast

52- The amniotic cavity develops:

- on the tenth day
- within the inner cell mass near the cytotrophoblast
- in extraembryonic mesoderm
- none of the above

53- "Quickening" or the period when fetal movements are commonly first felt by the mother occurs:

- near the end of the first trimester
- around the middle of the second trimester
- at the end of the second trimester
- . at the end of the embryonic period

54- Maternal serum alpha-fetoprotein increases if the fetus has

- oligohydramnios
- Bochdalek hernia
- . Ladd's bands
- anencephaly

55- Myelomeningocele (using clinical terminology)

- is most common at the tip of the coccyx
- leaves the affected individual with paralyzed upper limbs but able to walk
- is caused by a failure of the neural folds to meet and fuse
- is undetectable by the results of amniocentesis

56- A 4-year-old healthy female presents to your office to establish primary care. What phase of meiosis should her oocytes be in?

- Meiosis I - prophase
- Meiosis II - prophase
- Meiosis I - metaphase
- Meiosis II – metaphase

57- 22 Which of the following structures does not turn under onto the ventral surface of the embryo during folding of the head?

- heart
- notochord
- pericardial cavity
- septum transversum

58- The human placenta

- has a maternal component formed by the decidua capsularis
- shows no changes with age
- retains the cytotrophoblast layer of the placental barrier throughout gestation
- has chorionic villi as the structural and functional unit

physiology

59- A motor unit consists of

- An axonal terminal and the group of skeletal muscle fibers it innervates.
- An axonal terminal and whole muscle it innervates.
- A motor neuron and the group of skeletal muscle fibers it innervates.
- A motor neuron and whole muscle it innervates.

60- Which of the following is believed to initiate the contraction process in the skeletal muscle fibers?

- Actin-myosin attachment
- The strong affinity of the troponin C for calcium ions
- Sliding filament mechanism
- Adenosine triphosphatase activity of the myosin head

61- Compared to the skeletal muscle fibers, what is the cause of slow cycling of the myosin cross-bridges in the smooth muscle cells?

- Far less ATPase activity of smooth muscle fibers
- Response to different types of stimuli by smooth muscle fibers
- Higher activity of myosine kinase
- Low affinity to Ca ions in smooth muscle fibers

62- In which of the following muscle, control is exerted mainly by nerve signals?

- Ureters
- Ciliary Muscle
- Bile Duct
- Uterus

63- Which of the following is not matched the excitation contraction coupling in the skeletal muscles?

- Transverse tubules
- Multiple adherent points
- Ryanodine receptors
- Dihydropyridine receptors

64-



What does this figure show?

- Stress relaxation phenomenon
- Smooth muscle contraction
- Latch mechanism
- Smooth muscle relaxation

65- In an electromyography report, findings are compatible with severe axonal lesion of a nerve supply to a skeletal muscle. What symptom do you expect to observe in this patient?

- Hyperplasia of muscle fibers
- Muscle fatigue
- Atrophic changes
- Remodeling of muscle fibers

66- Which of the following is correct about isometric contraction?

- The tension on the muscle remains constant during the contraction.
- The load can not be lifted.
- It occurs when the force of the muscle contraction is greater than the load.
- The muscle shortens.

67- Which of the following best refers to the importance of chemical synapses?

- Synchronous firing of interconnected cells
- Signals amplification
- Rapid firing
- fast conduction velocity

68- Action potentials spread to the interior of the skeletal muscle fiber by way of

- transverse tubules
- terminal cisternae
- longitudinal tubules
- endoplasmic reticulum

69- Which of the following represents one of the electrical synapses properties?

- Presynaptic vesicles
- Synaptic delay
- Chemical transmitter
- Bidirectional transmission

70-  In this figure, what does this sign shows?



Acetylcholine molecules

Acetylcholine estrase

Ca ions

Dens bar

71- Which factor inhibits the active sites of actin filaments?

Inactivation of myosin filaments

ATP binding to actin filaments

Myosin heads detachment

Tropomyosin-troponin complex

72- Which of the following best describes the significant difference between skeletal muscle and large nerve fibers?

Velocity of conduction

Generation of action potential

Voltage gated sodium channels presense

The amount of resting membrane potential

73- Which of the following represents "Acetylcholine gated channels"?



D

B

A

C

74- During rigor mortis, contraction cycle is stopped at step 3 because of

lack of ATP

low affinity of actin filaments to myosin heads

non-functional heads of myosin filaments

ADP and Pi release

75- In this type of muscle, the source of Ca^{2+} ions needed for contraction is just from sarcoplasmic reticulum?

Biceps muscle

Iris muscle of the eye

Blood vessels

Uterus